



FABRIC MANUFACTURING

Part: B

Course Code: FE 253-0723

Weft Knitted Structure

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CLASSIFICATIONS OF WEFT KNITTED FABRIC

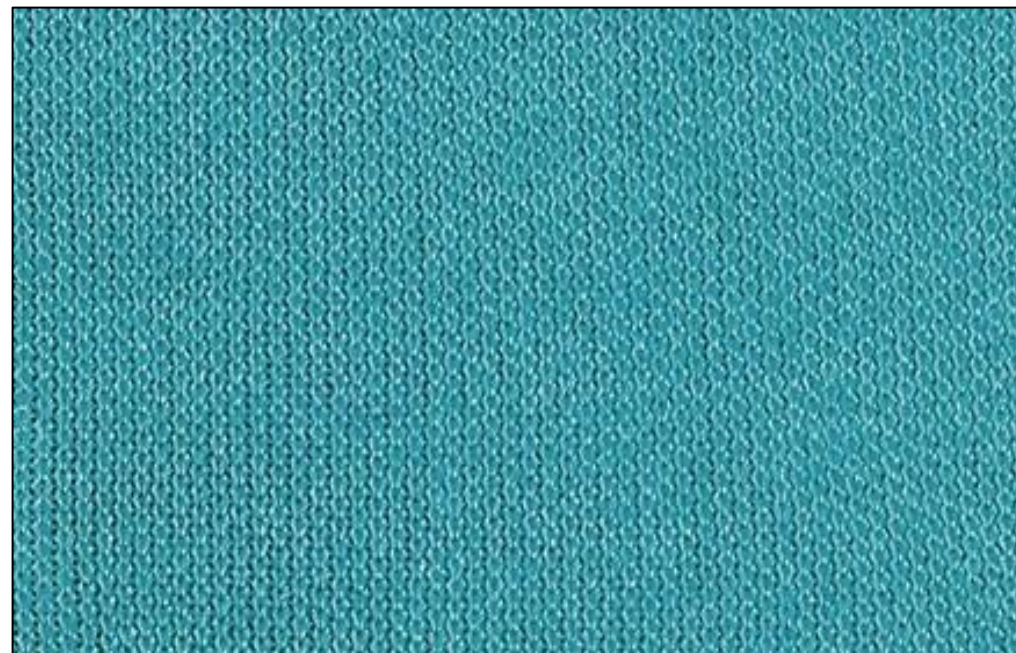
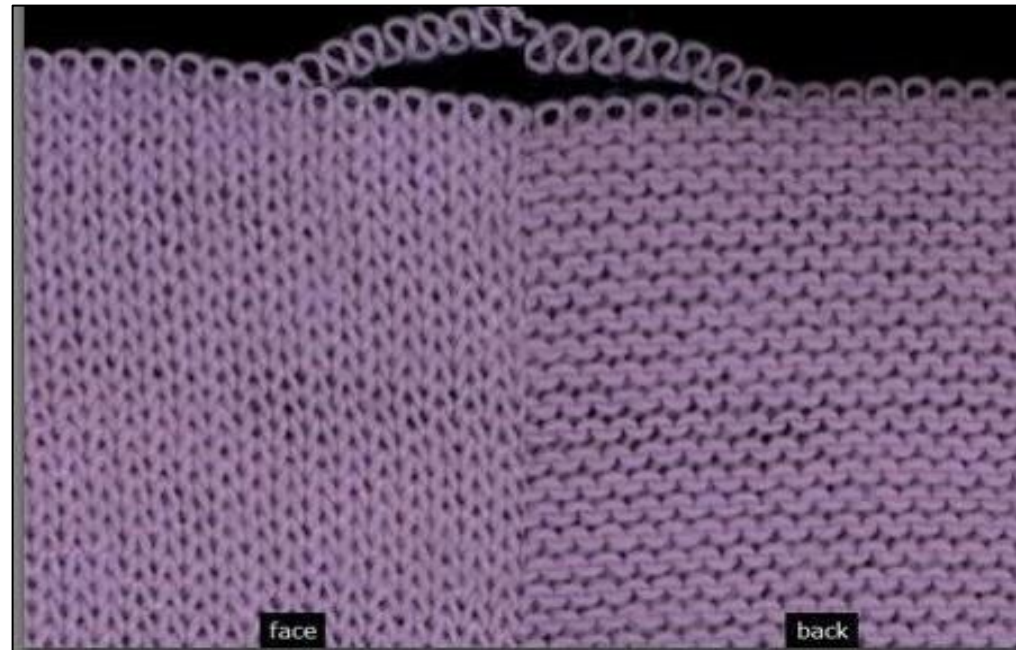
A. Single Jersey Fabric

B. Double Jersey Fabric

1. Rib

2. Interlock

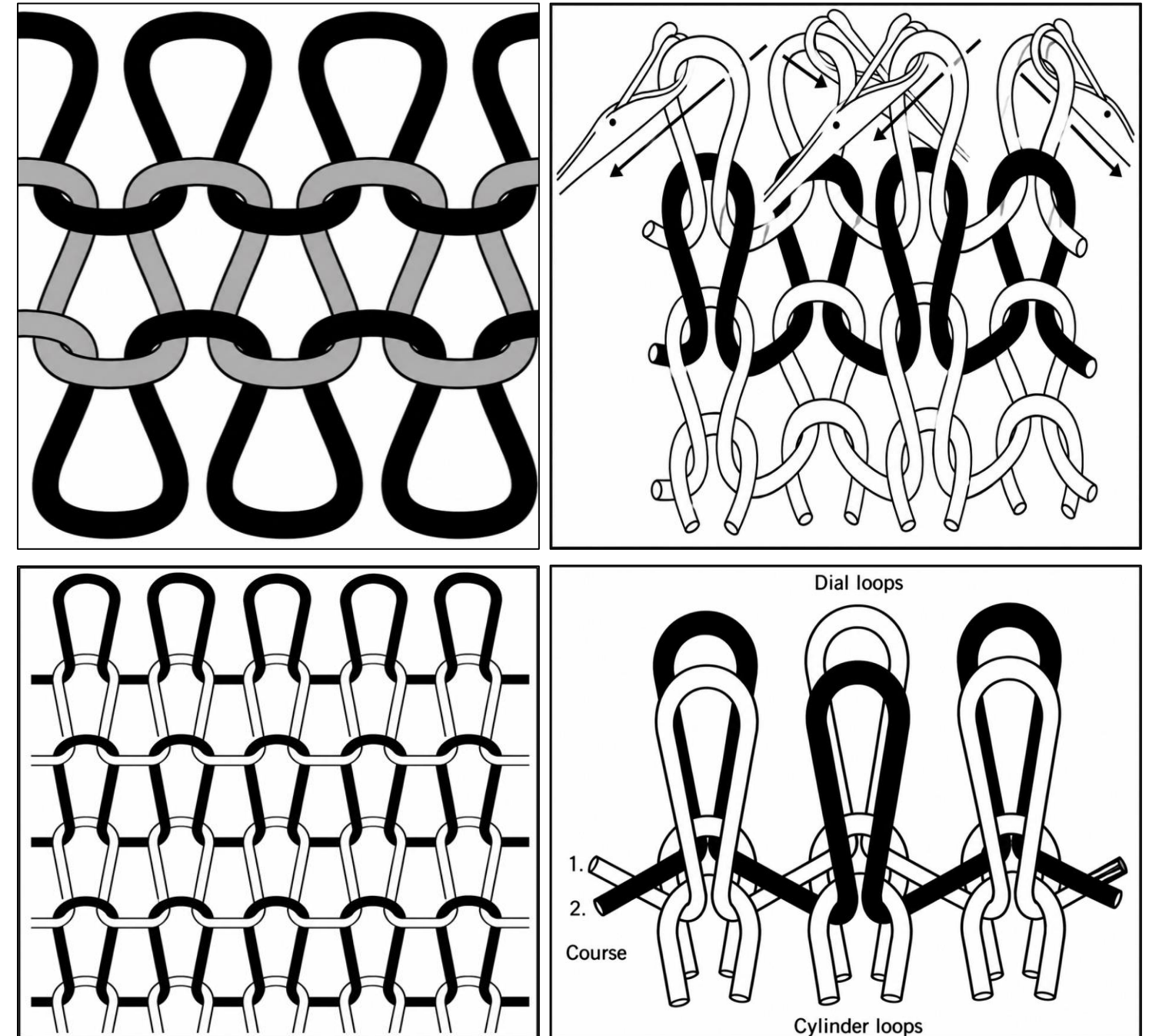
3. Purl



CLASSIFICATIONS OF BASIC WEFT KNITTED FABRIC

Basic weft knitted structure:

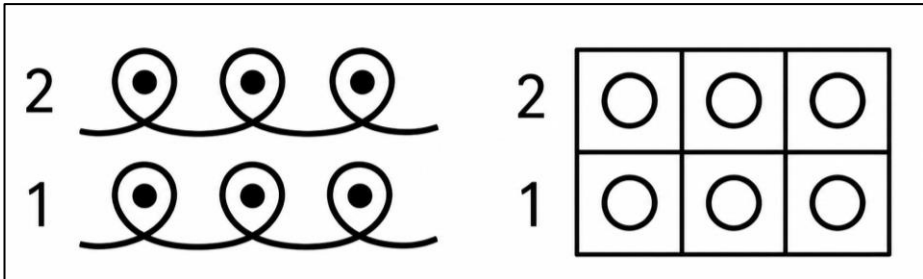
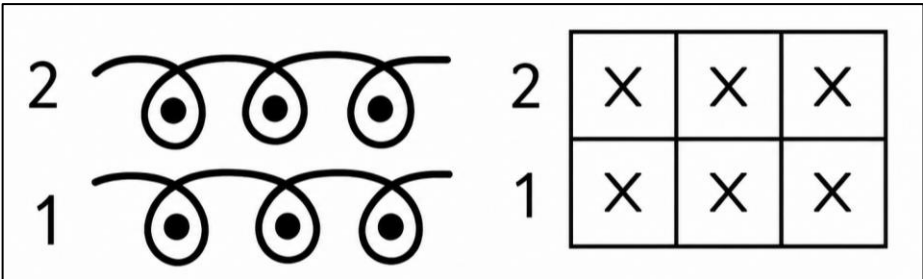
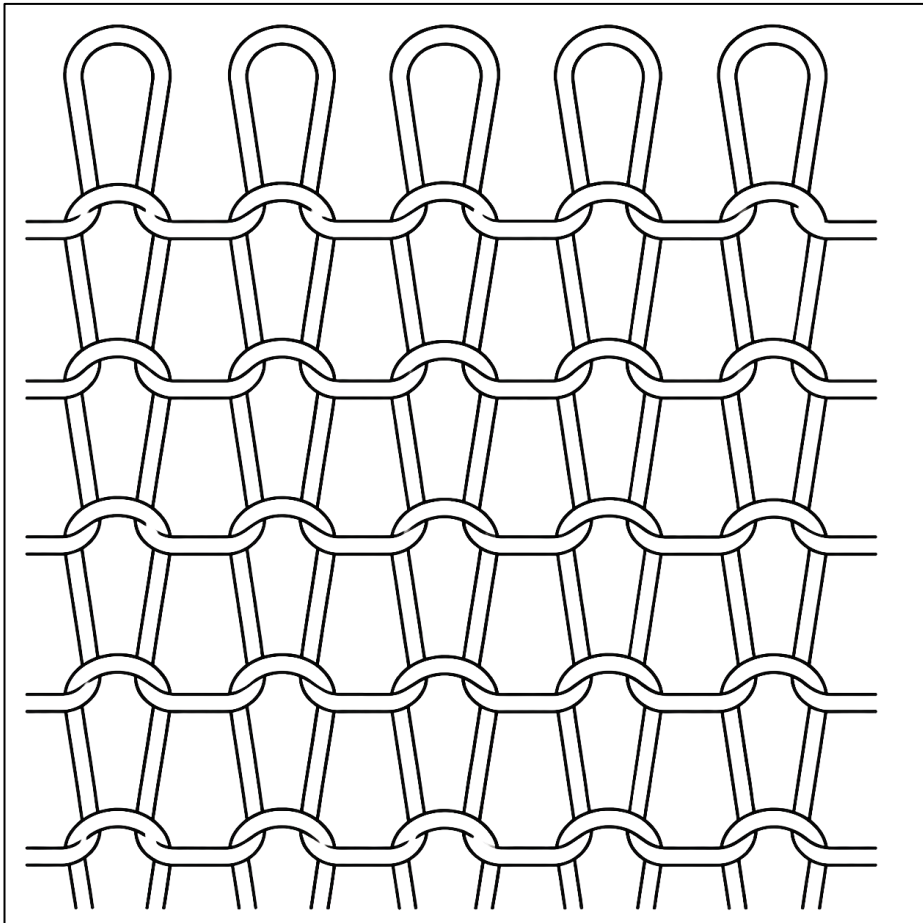
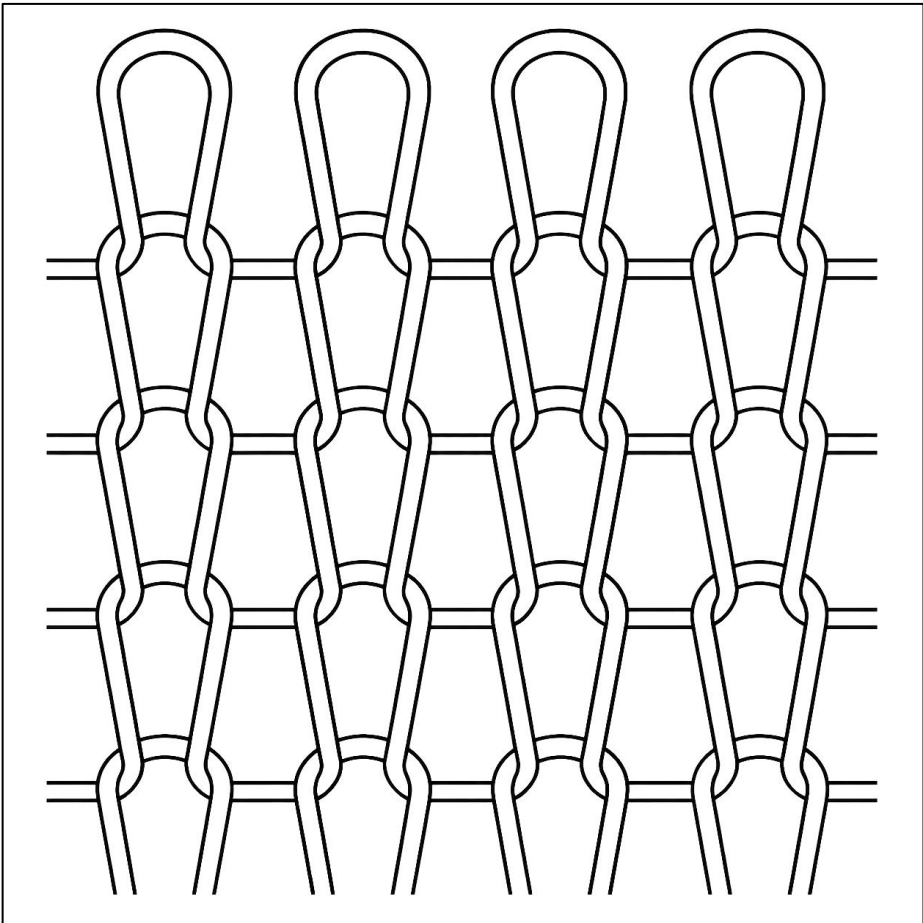
1. Single Jersey Plain
2. 1x1 Rib
3. 1x1 Interlock
4. 1x1 Purl



FEATURES OF SJ PLAIN FABRIC

- Single jersey plain fabric is simple, most common and economical.
- Both sides **are not identical**. Each side of the fabric is made of a single type of loop i.e. either face or back (reverse).
- The loops have a V-shaped appearance on the technical face and semi-circular on the technical back of the fabric.
- The V-shaped stitches on the face gives the smooth surface when compared with the back.
- Its **production rate is very high** because of stitch simplicity.
- The amount of elasticity varies according to the yarn used and the stitch length of yarn. The higher the **stitch length**, the more elastic the structure will be and simultaneously the poorer the bursting strength.
- It can be **pulled out freely from either end**.

DRAW THE LOOPING DIAGRAM OF TECHNICAL FACE AND TECHNICAL BACK OF A PLAIN FABRIC WITH NOTATION



HOW TO DISTINGUISH A SINGLE JERSEY (SJ) FABRIC?

- Appearance: Different on Face and Back. V shapes on face & arcs on back.
- Unroving: Either end.
- Curling: Tendency to curl.
- Thickness & Warmth: Thicker and warmer than plain woven fabric made of same yarn.
- Extensibility (L-W-A): Moderate – High – Mod-High.
- End uses: Ladies stockings, Fine cardigans, Men's and ladies' shirts, Dresses.

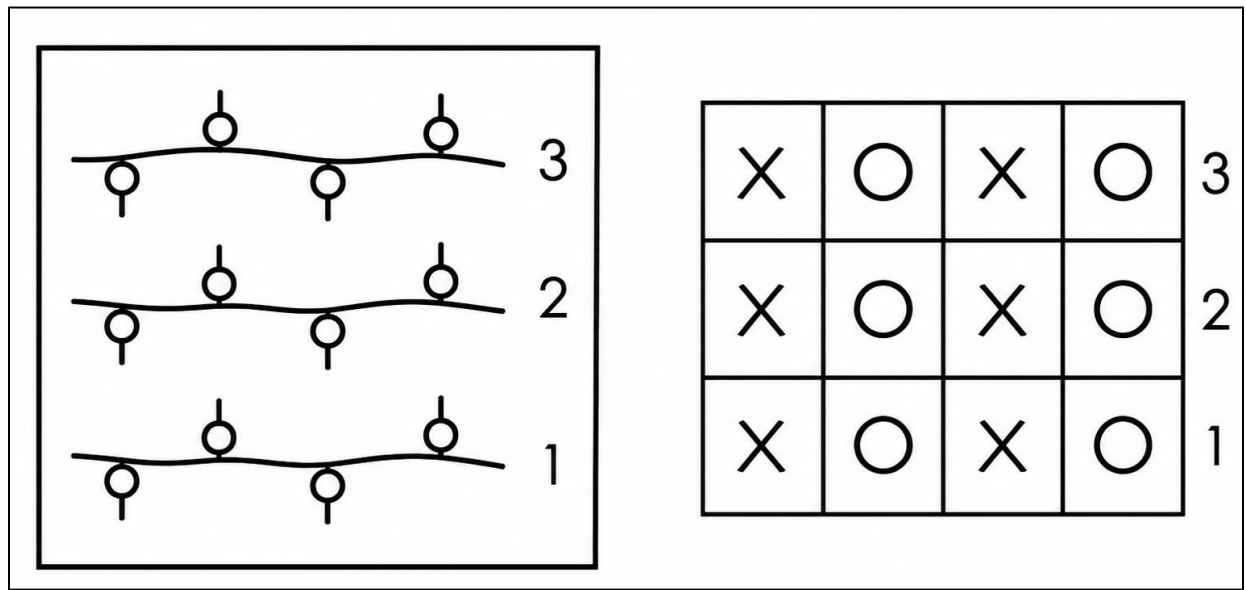
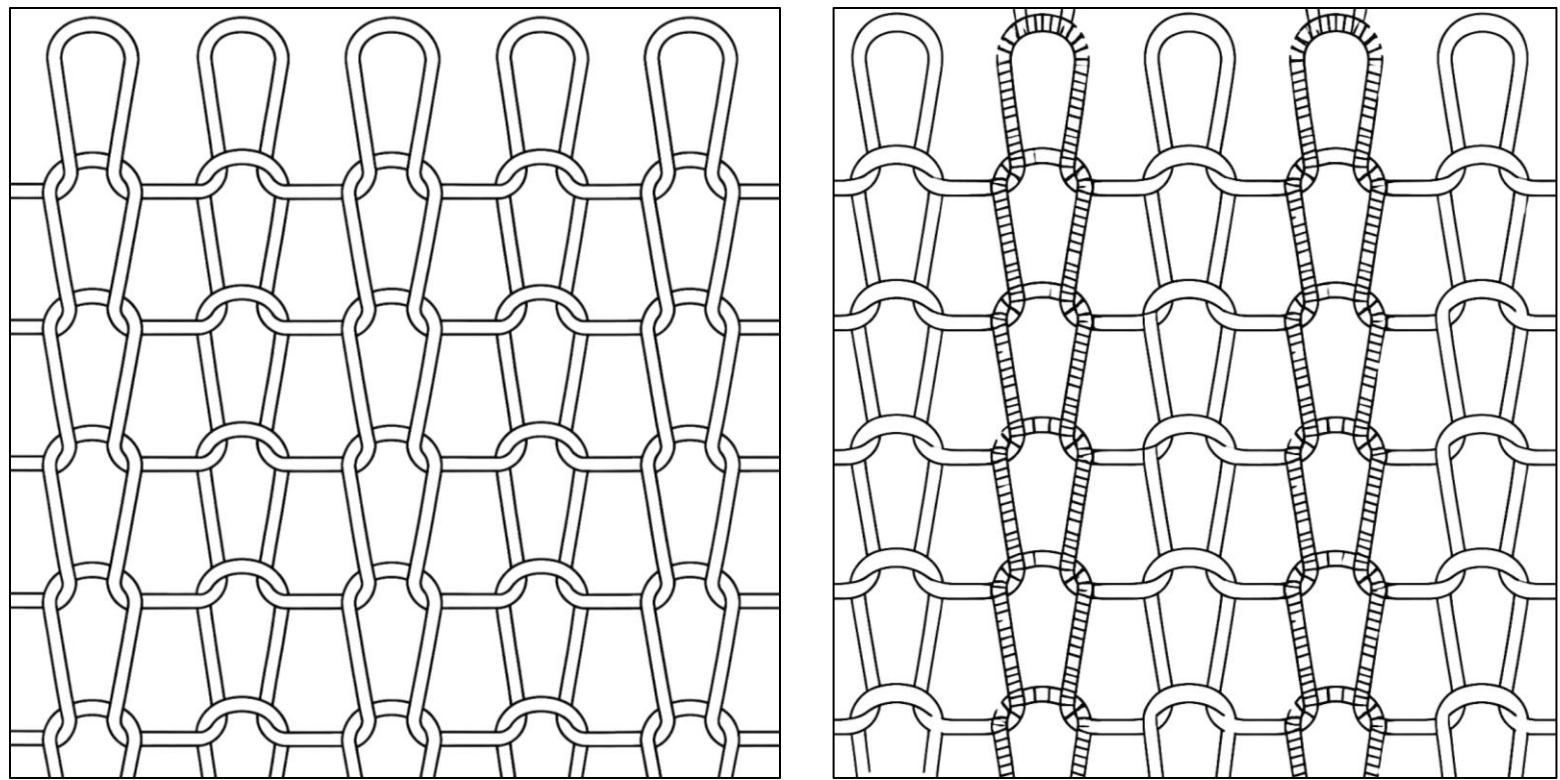
FEATURES OF RIB FABRIC

- The simplest rib fabric is 1x1 rib.
- It is knitted with two sets of single ended latch needles.
- Face and back of 1x1 rib are identical. Wales (technical face) prominent on the both sides and the structure can be named as double face or double-knit structure. But drop needle structures (2x2, 3x2, 5x4 etc.) are different.
- The fabric being in many cases symmetrical on both faces is not exposed to unbalanced stress and therefore does not curl, it lies flat, when cut.
- Rib machine requires finer yarn than similar gauge plain machine. But It is heavier structure than plain fabric.
- It is more expensive fabric to produce than plain.
- 1x1 rib is balanced by alternate wale of face loops on both sides. In face side, 1 wales face and another wales back. In back side, 1 wale back, another wale face. So, both face and back loops stay in the same course.
- Both side of the fabric imitate face side of single jersey as back loops remain hidden inside the structure. But back loops can be seen when stretched. A face wales are made by the needles of the cylinder, and the back wales are made by the needles of the dial.

FEATURES OF RIB FABRIC

- Rib structures have **alternate gaiting**.
- The 1×1 rib fabric is very elastic and can return to its original width after stretching. This happens because the face loop wales move over the reverse loop wales, allowing the fabric to recover easily. Rib fabric is very flexible and spring-like in the width direction because the yarn tries to return to its original shape. Due to this elasticity, rib structures are commonly used for elastic bands in garments, such as cuffs, waistbands, and collars. In normal constructions, rib fabric can stretch up to 120% along the course direction. In the wale direction, rib fabric behaves like plain fabric and has very limited stretchability.
- As the number of wales in each rib increases, the elasticity decreases because the number of changeovers from back to front diminishes. **Over 3x3 rib the fabric more and more behaves like plain fabric.**
- It cannot be pulled out freely from either ends. It can only be pulled out from edge knitted last. Rib structures, however, cannot be unravelled from the edge knitted first i.e. from the bottom. The arms of the connecting loops enter the loops above them from both sides. Any attempt to pull the yarn causes the connecting loops to tighten and blocks unravelling. Because rib cannot be unravelled from the end knitted first and because of their elasticity, they are suited to the edge of garments such as the tops of socks, cuffs and the waist edge of garments.

**DRAW THE LOOPING DIAGRAM OF TECHNICAL FACE AND
TECHNICAL BACK OF A RIB FABRIC WITH NOTATION**



HOW TO DISTINGUISH A RIB FABRIC?

- Appearance: Same on both sides. Like face of plain.
- Unroving: Only from end knitted last.
- Curling: No tendency to curl.
- Thickness & Warmth: Much thicker and warmer than plain woven fabric made of same yarn.
- Extensibility (L-W-A): Moderate – Very High – High.
- End uses: Socks, Cuffs, Waistbands, Collar, Men's outerwear, Underwear.

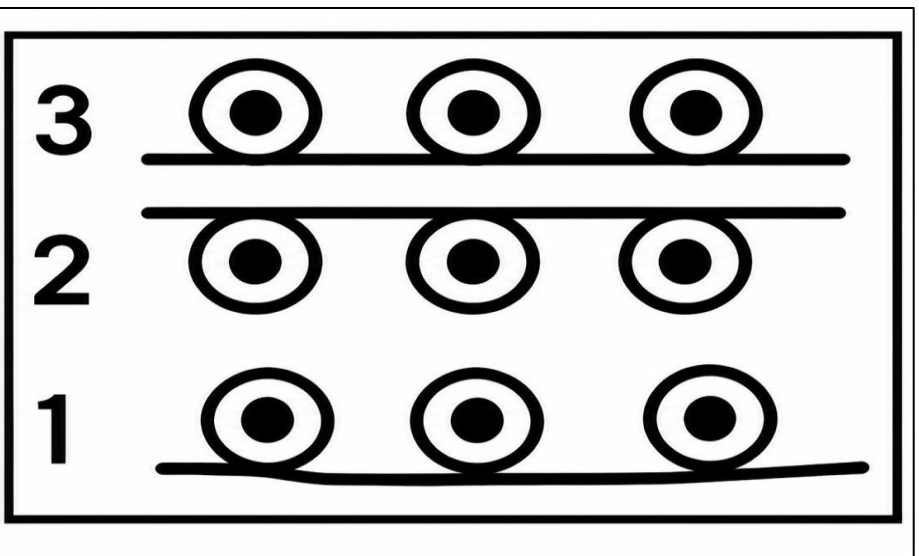
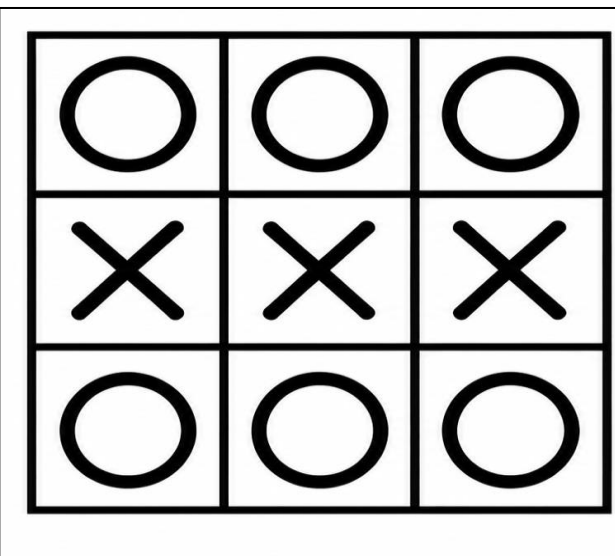
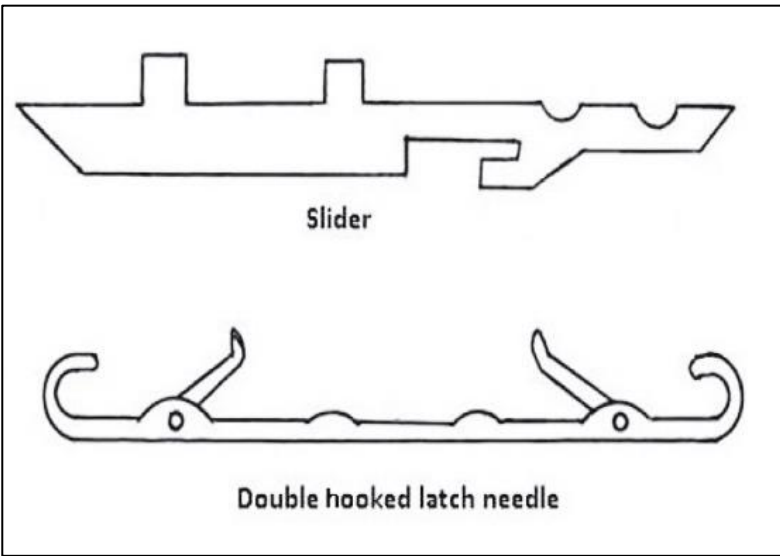
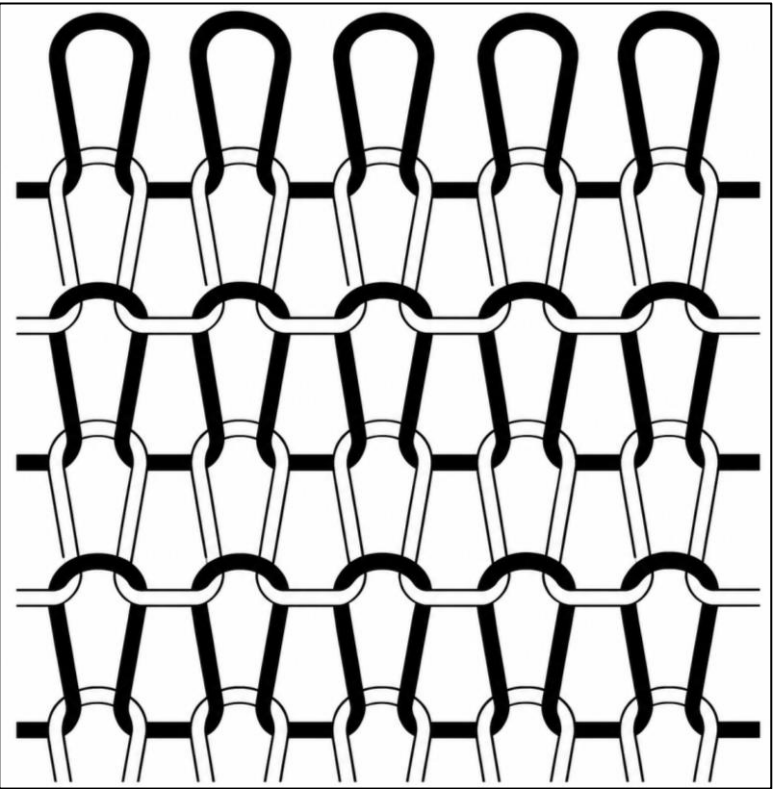
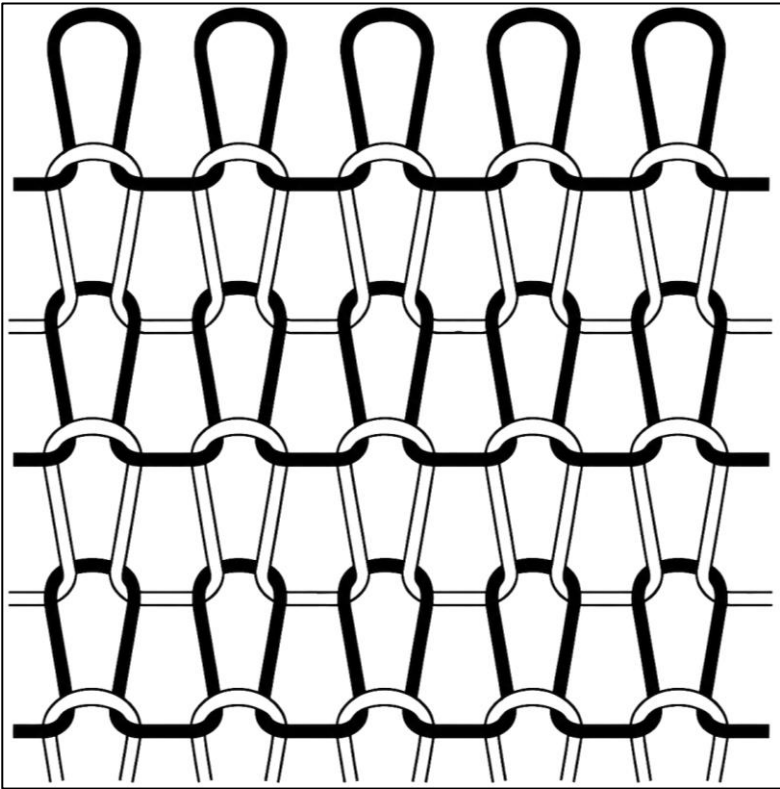
FEATURES OF PURL FABRIC

- Purl fabric is composed of sequentially 1 course back and 1 course wales that a horizontal or course-wise effect is produced.
- Both face and back loops remain in the same wales. The face loops are covered so the needle and connection arches typical of reverse loops show on both sides. So, the fabric imitates back side of single jersey.
- Needle can change the needle bed in alternate cycle. Same needle produces face loop on one bed and back loop on other bed. Needle slider is used to move the needle set from one bed to another.
- This fabric has no tendency to curl when it is cut.
- It can be pulled out from either end.
- Double ended latch needles are used to produce purl fabric.
- Its thickness is theoretically double to that of a plain knit fabric.

FEATURES OF PURL FABRIC

- It is a balanced structure and has a soft handle.
- Purl structure creates more extensibility in length wise direction. This is an advantage, especially when compared with the very limited lengthwise extensibility of plain or rib structures. The fabric is stretchable in the width as with all loop-based structures.
- Purl knit structures are best suited to infant's wear, socks, golf sweaters and sportswear. This flexibility in length and width makes the purl knit ideal for baby wear where elongation and expansion are required due to the fast growing rate of infants and to simplify the dressing process.

DRAW THE LOOPING DIAGRAM OF A PURL FABRIC WITH NOTATION



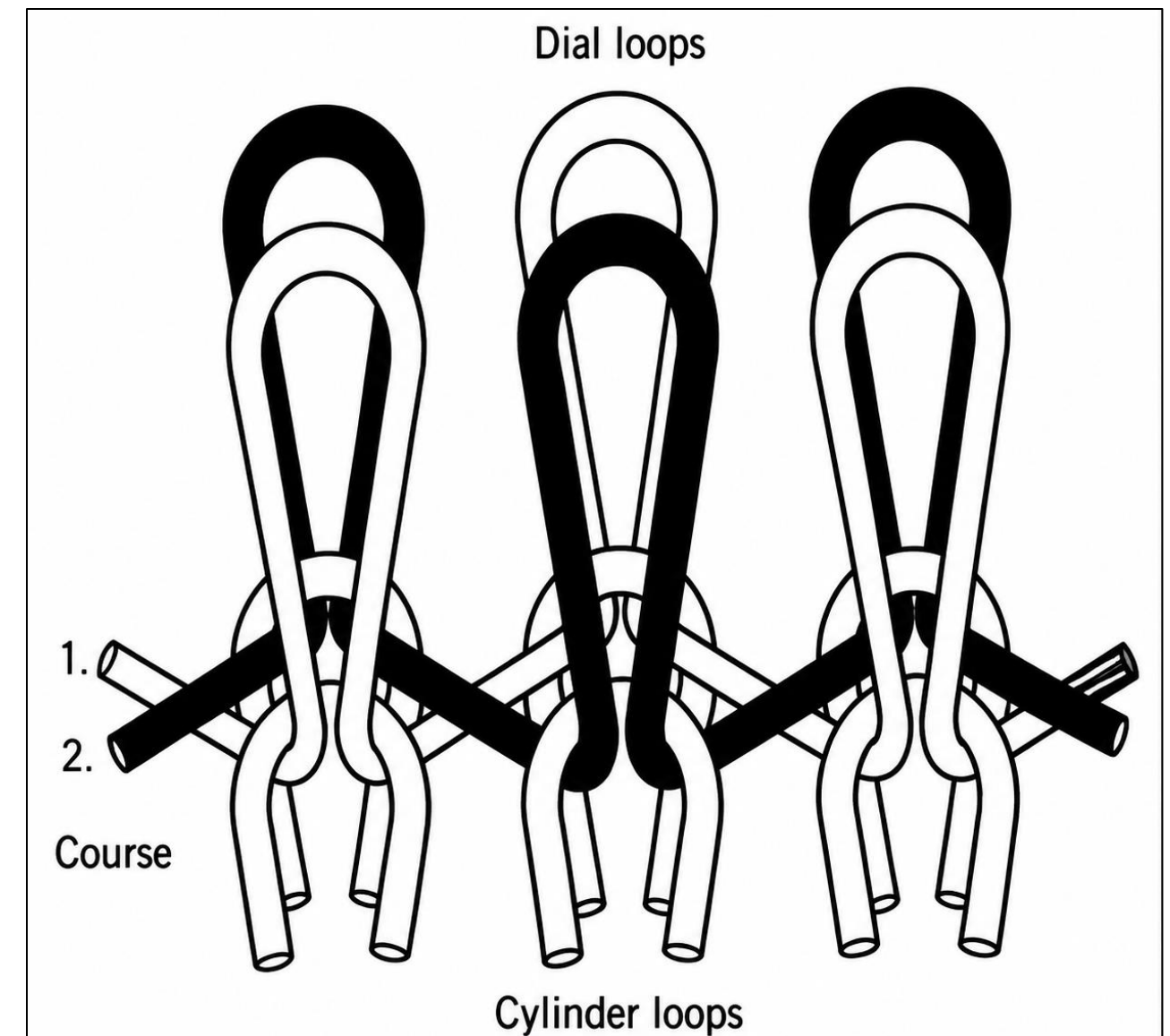
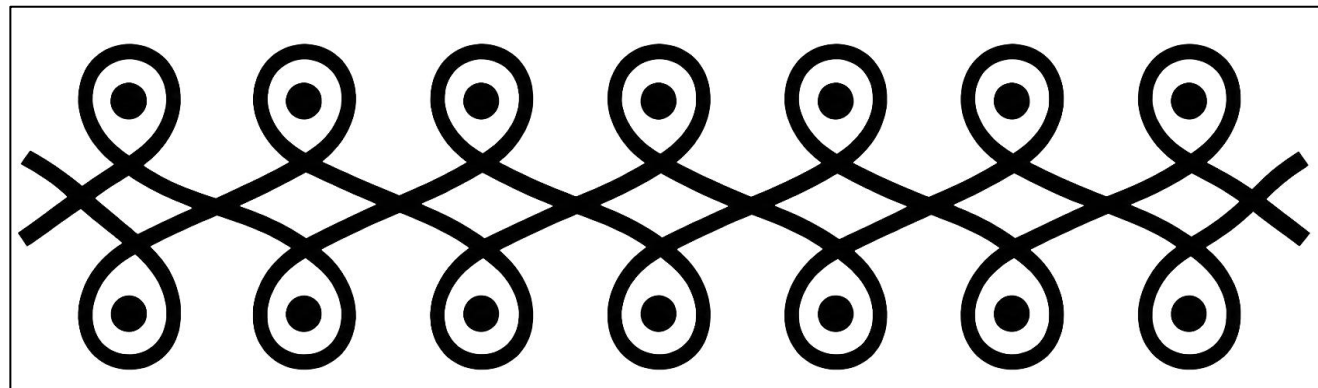
HOW TO DISTINGUISH A PURL FABRIC?

- Appearance: Same on both sides. Like back of plain.
- Unroving: Either end.
- Curling: No tendency to curl.
- Thickness & Warmth: Very much thicker and warmer than plain woven fabric made of same yarn.
- Extensibility (L-W-A): Very High – High – Very High.
- End uses: Children's clothing, Thick & heavy outdoor wear.

FEATURES OF INTERLOCK FABRIC

- Interlock has the **technical face of plain fabric on both sides**.
- It cannot be pulled out from either end. It can only be **pulled out from edge knitted last**.
- It is a balanced, smooth and stable structure. it is thicker, heavier and narrower than rib of equivalent gauge, and requires a finer, better, more expensive yarn
- **It lies flat without curl when cut.**
- Face and back of plain interlock fabrics are same. Wales prominent on the both sides. The fabric imitates face side of single jersey fabric.
- The needles in two beds being exactly opposite each other (**Face to face gaiting**) so that only one of two can knit at one feed. The wales on each side are exactly opposite each other.
- Each interlock pattern row or interlock course requires two feeder courses. So, 1 x 1 interlock structure is produced by **two 1 x 1 rib structure interlace to each other**. Productivity is half of rib knitting machine.
- Interlocks fabrics are **thicker and heavier than rib fabrics**.

DRAW THE LOOPING DIAGRAM OF AN INTERLOCK FABRIC WITH NOTATION



HOW TO DISTINGUISH AN INTERLOCK FABRIC?

- Appearance: Same on both sides. Like face of plain.
- Unroving: Only from end knitted last.
- Curling: No tendency to curl.
- Thickness & Warmth: Very much thicker and warmer than plain woven fabric made of same yarn.
- Extensibility (L-W-A): Moderate – Moderate – Moderate.
- End uses: Underwear, Shirts, Suits, Trouser, Sportswear.

THANK
YOU !

